

Acoustic-Video Perception for Robot Manipulation with Occluded Contact Dynamics

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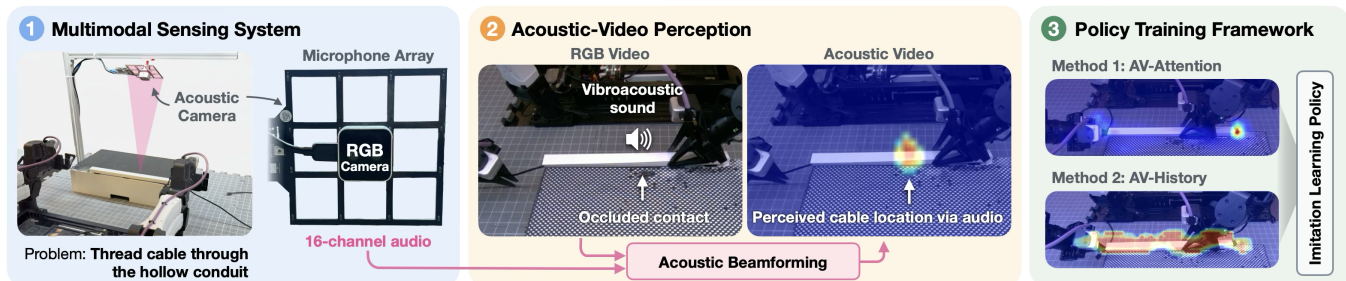


Fig. 1. Overview of the Acoustic Video (AV) robot perception system. The overhead acoustic camera localizes the sound source on the RGB camera plane. We integrate this modality into visuomotor policies in two ways: (1) AV-guided attention supervision and (2) AV-history auxiliary observation.

Abstract—Almost all manipulation tasks involve occluded contact dynamics: it is inherently difficult to visually observe detailed surface contacts as they are created and changed. Consider grasping, assembly, peeling, or pushing – even a planar task like “Push T” hides the contact between object and surface. In some cases, occluded contact generates perceptible *vibroacoustic* sound through friction and provides useful contact information. In this paper, we explore how a low-cost “acoustic camera” can perceive occluded contact dynamics to enhance robot manipulation. We use microphone array beamforming to localize audio over time in a 2D RGB camera image plane. When overlaid on video, we call this *Acoustic Video*. We present two approaches: 1) AV-Attention and 2) AV-History. We build both AV approaches on the encoder-only Action Chunking Transformer for an efficient manipulation policy trained by imitation. Across five contact-rich manipulation tasks, experiments suggest that acoustic video can improve task success rate by 21.0% and progress by 16.1% on average compared to vision-only baseline and conventional mono-channel audio approach. Video figures are available at <https://rgba-robot.github.io/>.

I. INTRODUCTION

Humans and many animals use spatial audio perception [1]. Inspired by nature, audio has long been used in robotics: as early as 1989, a small autonomous robot called Squirt [2] used sound to guide its movement, mimicking the behavior of a bug that ventures out in the direction of noises. Microphone arrays¹ were developed in the early 20th century during World War I, originally to triangulate gun positions based on sound arrival times [3]. Since then, microphone arrays have also been used in robotics, such as drones for search-and-rescue tasks [4], [5], human-robot interaction (HRI) for conversational agents [6], [7], and manipulation using contact microphones [8]–[11]. Despite this history,

using spatial acoustics directly grounded in the visual scene for a robot manipulation policy is underexplored.

Many manipulation tasks involve occluded contact dynamics that are difficult to observe visually as they form and change. Consider threading a cable through a hollow conduit (Fig. 1), where the occluded endpoint location is critical for identifying the task stage. Even a planar task like “Push-T” depends on surface contact underneath the object, and the correct pushing direction depends on how contact evolves. These interactions, however, make friction between the contacting surfaces and radiate *vibroacoustic*² sound. This sound provides a useful contact cue, but prior approaches perceive it with a single-channel signal that carries no spatial information about the contact [12]–[19]. As a result, the policy must implicitly learn the relationship between acoustic and visual observations from inherently limited information.

This paper explores acoustic video perception for tasks involving *vibroacoustic* sound. The array microphone (*Acoustic Camera* [20]) can localize the sound in the scene during manipulation. We use acoustic beamforming at the target frequency associated with contact friction (10 kHz). We then create time-varying acoustic intensity maps overlaid on an RGB video stream, which we call *Acoustic Video* (AV).

We propose two methods to integrate this novel modality into imitation learning policies: 1) **AV-Attention** supervises the model’s visual attention mechanisms to focus computation on active contact regions. 2) **AV-History** incorporates acoustic video as a policy observation to track task progress over time. While the AV-History method uses live audio as a reactive input at deployment, the AV-Attention method acts purely as a training-time framework, allowing inference on standard vision inputs.

¹Paper under review.

²a set of two or more individual microphones placed in a specific geometric layout to capture sound from multiple points in space at the same time.

²sound radiated by a vibrating structure, here induced by friction at the contact.

We evaluate the potential of our Acoustic Video perception with these manipulation tasks as shown in Fig. 4: thread cable through a conduit, grind with a rotary tool, peel transparent protector film on smartphone, cut seal to decant box, and wipe blue line from product. Evaluations on five real vibroacoustic tasks suggest that acoustic-video perception can improve task success rate by up to 21.0% and task progress by 16.1% compared to models trained on vision only and a conventional mono-channel audio approach.

II. RELATED WORK

We first review prior work on multimodal perception for manipulation, and then discuss two topics more specific to our approach, spatial audio in robotics, and spatial or auxiliary supervision for robot learning.

A. Multimodal Perception for Robot Manipulation

In contact-rich manipulation, interaction state is often only partially observable from vision. Contact, friction, slip, and applied forces can be difficult to infer visually, particularly under occlusion or when manipulating transparent or specular objects. Prior work has therefore combined vision with complementary physical sensing, including force [21]–[23] and tactile [24]–[27] feedback, acoustics [18], [28], capacitive sensing [29]–[31], and imaging beyond the visible spectrum [32], [33].

Acoustic sensing is a promising modality, as microphones are relatively inexpensive and can also capture contact sound remotely, through air or a solid medium, beyond the region the sensor physically covers. Some approaches add active acoustic actuators and measure the acoustic response of a manipulating object [10], [34]. Many tasks, however, also produce sound passively, without any active component: tapping, wiping a surface, or pushing an object, all excite micro-friction vibrations that radiate as *vibroacoustic* sound. Prior work has explored this signal to improve manipulation under occlusion, but mostly consumes mono-channel audio as waveforms or spectrograms as input [12]–[19] – for example, learning contact-rich skills from in-the-wild audio-visual demonstrations [12], [13], acting under visual occlusion or for dynamic tasks [14]–[17], and detecting in-hand slip [18], [19]. While these works show that acoustic observations can reveal task-relevant interaction information difficult to infer from vision alone, mono-channel audio carries no spatial cues. However, in many tasks, such as threading a cable through a conduit to reroute it (Fig. 1), the location of the contact is important information the policy needs. It must therefore implicitly learn to bind the sound to a place in the visual scene, a mapping that is costly or sometimes infeasible to learn.

Our work follows these tasks with vibroacoustics but uses remote acoustic sensing rather than contact-mounted sensing. A microphone array observes interaction sounds without instrumenting the manipulated object or contact surface, while spatially grounding the acoustic signal in the visual scene. This allows fusing acoustic information with

visual features at corresponding image locations—a spatial correspondence that prior work has left underexploited.

B. Spatial Audio in Robotics

Humans and many animals use spatial audio perception to understand and navigate the world better [1]. In robotics, researchers have used spatial audio in mainly three different ways: ultrasonic ranging, source sound separation, and sound source localization.

Active acoustics (ultrasonic ranging) has a long history in mobile robotics, where early systems measured proximity to walls and obstacles for navigation by emitting a sound pulse and timing the returning echo [35], [36]. Unlike other time-of-flight sensors such as LiDAR, ultrasonic ranging reflects off transparent surfaces. Recent work fuses it with depth sensing to reconstruct glass walls via a learned implicit scene representation [37].

More related to our work, microphone arrays have been used to separate a target sound source from the noise. This line of research, often termed robot audition [6], [38], was motivated by human-robot interaction (HRI), where a robot must isolate and recognize a speaker from competing talkers or its own motor noise. Beamforming toward a known source direction with post-processing enabled real-time separation of simultaneous speakers on mobile robots [39], and relevant audio processing open-source packages for robotics have emerged [7], [40]. Commercial humanoids nowadays, such as Unitree G1 [41] and R1 [42] ship with built-in microphone arrays for voice interaction.

Most related to our work, microphone arrays have also been used to localize the sound source. With eight-microphone arrays, Valin et al. [43] implemented a mobile robot following moving speakers. Similarly, drones with an array have localized victims for search-and-rescue where vision is obstructed [4], [5]. More closely, in manipulation, a few works use contact microphones mounted in the end effector or arm to localize the contact or collision location [8], [9], [11]. These works, however, either localize the source in a direction-of-arrival (azimuth/elevation vector) or are bounded to the robot’s own body with contact microphones. None has considered localized sound in the visual task scene (i.e., beyond the robot body) for manipulation. In contrast, we ground the localized sound densely in the visual scene, aligning acoustics with the corresponding visual features, so that a learning-based manipulation policy can directly exploit where contact occurs.

C. Spatial and Auxiliary Supervision for Robot Learning

Robot policies are commonly trained end-to-end from demonstrations [44] or task rewards, but a growing body of work has shown that explicitly providing task-relevant spatial structure can improve policy learning and generalization. Human gaze and eye-tracking signals have been used to direct visuomotor representations toward regions relevant for manipulation [45], [46], and more recently as auxiliary regularization for imitation learning and vision-language-action policies [47]–[49]. Beyond gaze, prior work has

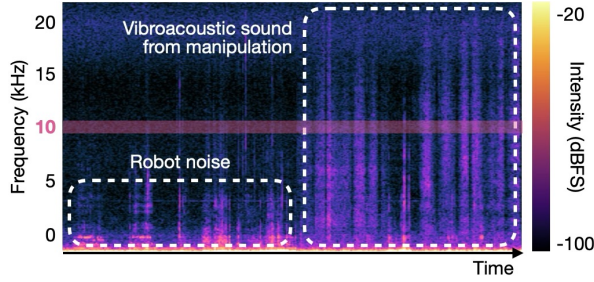


Fig. 2. Overall acoustic profile from manipulation vs. robot noise. The frequency 10 kHz is used for acoustic beamforming.

introduced spatial supervision through language-conditioned image keypoints [50], pixel-aligned contact representations [51], image-space affordance predictions [52], and auxiliary objectives that encourage policy attention to capture object grounding, spatial geometry, and temporal task structure [53].

These works demonstrate the benefit of explicitly exposing task-relevant spatial structure rather than expecting implicit emergence solely from action supervision. Our work similarly provides spatial supervision, but derives it directly from physical interaction rather than from human gaze, manually specified regions, or visually inferred affordances. We localize interaction sounds in the camera frame and use the resulting acoustic video to supervise policy attention. This provides a physically grounded signal of where task-relevant interaction occurs and enables acoustic information to either shape the learned visual representation or directly inform the policy when visual observations alone may be ambiguous.

III. PROBLEM SETUP

We consider contact-rich manipulation tasks whose contact dynamics are occluded from vision; for example, hidden behind the tool or gripper, enclosed within an object, or made with a transparent or specular material. Many such tasks are also vibroacoustic, passively generating sound as contact friction is created and changed. Our goal is to learn a visuo-acoustic policy from demonstrations that exploits this sound by localizing the sound in the image frame that explicitly aligns the acoustic signal with the corresponding visual features. We assume a static environment in which the manipulated object does not substantially shift its position relative to the acoustic camera. We also assume a relatively consistent acoustic background (e.g., HVAC or AC-driven lighting) without abrupt or transient noise.

IV. ACOUSTIC-VIDEO PERCEPTION SYSTEM

In this section, we describe the hardware and software pipeline of the proposed acoustic-video perception system.

A. Acoustic Camera Hardware

To capture spatially grounded visuo-acoustic observations, we implemented an overhead acoustic camera with two commercial components: a 16-channel microphone array (miniDSP UMA-16 v2 [54]) and an RGB camera (Intel

RealSense D405 camera [55]). The camera is mounted in the center of the array, as shown in Fig. 1. The microphone streams synchronized 16-channel audio sampled at 44.1 kHz over a USB 3.0 interface. The RGB camera records an overhead image at a resolution of 1280×720 at 30 Hz. We spatially calibrate the array’s computed sound-pressure maps to match the camera’s native field of view (89.0° horizontal $\times$ 57.9° vertical). To this end, we define the beamforming search grid as a planar region at a fixed depth D from the array, sized to span the camera’s field of view at that depth. Given the horizontal and vertical fields of view, the dimensions of the grid are $W = 2D \tan\left(\frac{89.0^\circ}{2}\right)$, $H = 2D \tan\left(\frac{57.9^\circ}{2}\right)$ with a 2 cm spacing, so the beamforming map is directly mapped onto the RGB image frame.

B. Acoustic Beamforming

Fig. 1 describes the pipeline for creating AV observations. Consider a friction sound produced when the cable moves through a hollow conduit. This sound propagates to each of 16 channels at slightly different times due to their spatial disparity. For every candidate location $\mathbf{s}_{xy}$ on the planar search grid defined in Sec. IV-A, we first compute the distance $\|\mathbf{s}_{xy} - \mathbf{m}_i\|$ from the candidate source to each of the 16 microphones $\mathbf{m}_i$. Dividing by the speed of sound c gives the propagation delay $\frac{\|\mathbf{s}_{xy} - \mathbf{m}_i\|}{c}$ that the audio would have accumulated on channel i if it were originated from $\mathbf{s}_g$. We then realign the 16-channel audio by these delays and sum them. The construction from the source would constructively interfere to amplify the signal, and others would destructively interfere and cancel out noise.

In this process, we specifically combine Multiple Signal Classification (MUSIC) [56] and Steered Response Power (SRP) [57] in a complementary way. We use SRP on broadband audio for a robust yet spatially diffuse sound pressure map. We then use MUSIC at the target frequency for tight sound localization.

As shown in Fig. 2, contact dynamics during manipulation produce high-frequency acoustic signatures, exceeding 10 kHz. This distinct frequency profile enables separation of task-relevant contact audio from low-frequency robot motor noise (0–4 kHz) and mechanical vibrations or ambient (roomtone) sound (< 100 Hz). We therefore high-pass the raw signal at 3 kHz before feeding the broadband audio into SRP. For MUSIC, we select a single narrowband target at 10 kHz with a 517 Hz bandwidth. In principle, the MUSIC target frequency can be adapted online or use multiple targets, but we keep it fixed to minimize computation for real-time throughput. We threshold the SRP map at 90-th percentile to form a spatial gate and multiply it element-wise with the MUSIC map to get the sound pressure map, $A(x, y, t)$.

C. Acoustic-Video Perception Input

We define two temporally accumulated observations of sound-pressure maps: History and Future representations.

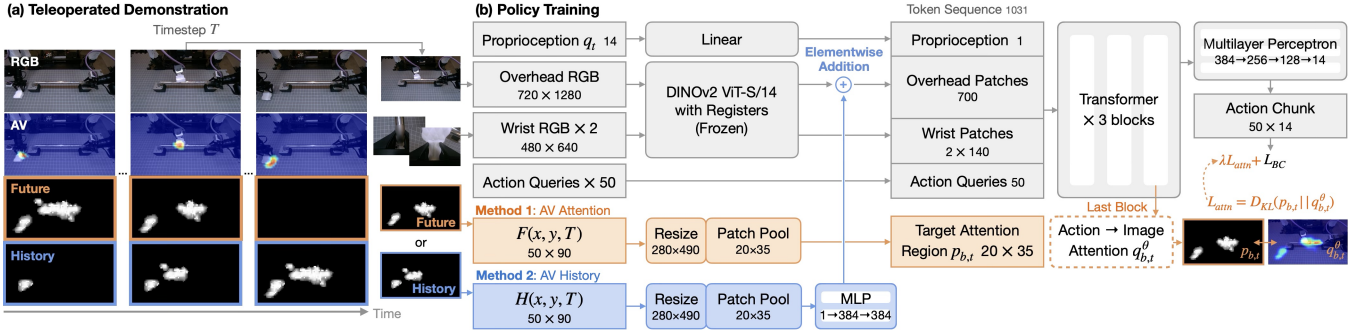


Fig. 3. Each frame of the demo episode includes three camera frames (1 overhead, 2 wrists) and an AV observation that constructs the input of each method. A path with orange color indicates method 1, **AV-Attention**: a target attention region ($p_{b,t}$) is used to compute the Kullback-Leibler (KL) divergence loss ($\mathcal{L}_{attn}$) with transformer attention ($q_{b,t}^\theta$). Then this loss is added to the behavior cloning loss ($\mathcal{L}_{BC}$) to update the model weights. Note that this attention supervision is done only on the overhead RGB frame. The path with blue indicates method 2, **AV-History**: a two-layer multilayer perceptron (MLP) embeds the per-patch AV-history scalar into a feature space, and the result is added element-wise to the overhead patches.

In the History representation $H(x, y, T)$, we max-pool all past sound-pressure maps up to frame t to track cumulative task progression, i.e., $H(x, y, T) = \max_{0 \leq t \leq T} A(x, y, t)$, where T is the elapsed time. Semantically, it is a log of surface areas that have been wiped in Fig. 3a). This representation is provided as a policy observation in Method 2, AV-History.

The Future representation $F(x, y, T)$ is the opposite of it – it max-pools remaining acoustic signals from frame T onward, i.e., $F(x, y, T) = \max_{t \leq T \leq T_{final}} A(x, y, t)$, where T is the elapsed time and T_{final} is the demonstration time. Semantically, it is the remaining target area to be further wiped in Fig. 3a). This is computed offline only in training demo episodes. This target supervises the visual attention maps in Method 1, AV-Attention.

V. ACOUSTIC-VIDEO POLICY LEARNING FRAMEWORK

We propose two policy learning frameworks to use AV perception: AV-Attention and AV-History. We first describe the base visuomotor policy architecture that both frameworks share, then explain each framework on this base architecture.

A. Base Visuomotor Policy Architecture

To use AV for robot control, we consider the Action Chunking Transformer (ACT) [58] with an encoder-only design for its efficient performance on a small amount of demonstrations. The gray-colored components in Fig. 3b illustrate the base policy architecture. We replaced ACT’s ResNet vision encoder with a frozen DINOv2 with registers [59] to better interpret the effects of attention supervision from Method 1 without high-norm artifact patches. All camera frames are encoded by this vision encoder, yielding 700 patch tokens (20×35) for the overhead view and 140 (10×14) for each wrist view. The 14-DoF bimanual joint configuration is embedded into a single proprioceptive token by an MLP, and the action chunk is represented by 50 learned action queries. We concatenate all these tokens in a single sequence (1031 length $\times$ 384 dimension) and process it with one stack of 3 self-attention blocks with 12 heads each. We intentionally make our architecture shallow to avoid overfitting on the limited demonstration data while retaining enough attention heads to fuse the multi-view and AV tokens.

We use the same action chunk size (1-second) as the original ACT; a three-layer multilayer perceptron (MLP) decodes the hidden states into a chunk of 50 joint displacements relative to input proprioception.

B. Method 1: AV-Attention

In our first method, we use acoustic-video perception to guide the imitation learning policy to attend to visual features directly related to the manipulation task. Following the convention of GuidedVLA [53] and Gaze-Regularized VLA [48], we add a training-only regularization term to the original behavior cloning loss. The orange-colored components in Fig. 3 show the implementation of this. Specifically, we update the model weights through the loss:

$$\mathcal{L}(t) = \mathcal{L}_{BC}(t) + \lambda \mathcal{L}_{attn}(t), \quad \lambda = 0.05 \quad (1)$$

$$\mathcal{L}_{BC}(t) = \|a_{gt}^{(t)} - a_\theta^{(t)}\|_1 \quad (2)$$

$$\mathcal{L}_{attn}(t) = \frac{1}{|\mathcal{G}_t|} \sum_{(b,t) \in \mathcal{G}} D_{KL}(p_{b,t} \| q_{b,t}^\theta) \quad (3)$$

where $p_{b,t}$ is the future-accumulated sound pressure map (Fig. 3a Future) pooled onto the patch grid, and $q_{b,t}^\theta$ is the attention from the action query to the image patches in the final transformer block, averaged over heads. $\mathcal{G}$ excludes observations whose target map is empty, i.e., the silent tail after the last acoustic event. We deliberately chose a small regularization weight of $\lambda=0.05$ per the findings of Pani and Yang [48], which describes that maintaining low regularization strength benefits from the statistical tendencies of attention supervision labels while preserving flexibility to discover task-optimal patterns that may occasionally deviate from the label. This attention supervision is applied exclusively to the overhead camera stream. While wrist views are not directly supervised, all camera views process tokens through the same shared attention heads. Crucially, Method 1 requires acoustic sensing only during training. In deployment, the policy operates on standard vision-only observations without requiring an acoustic camera or audio processing.

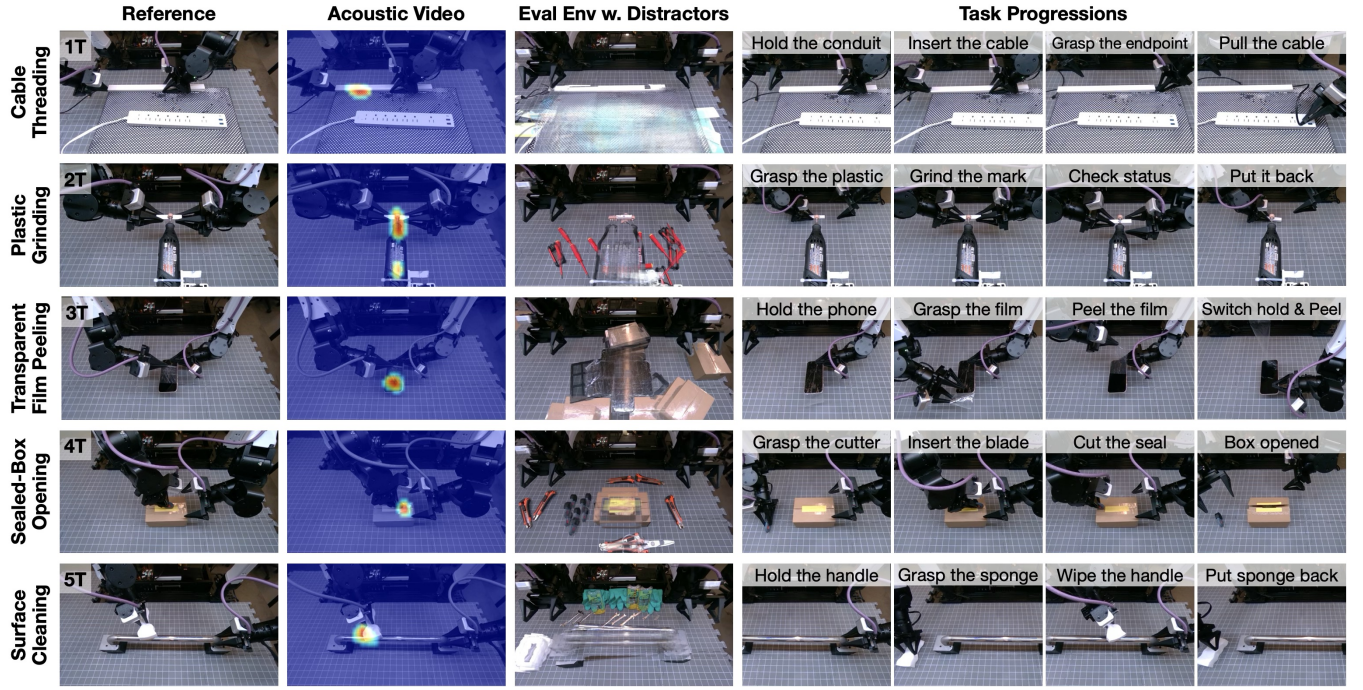


Fig. 4. Five real-world environments of our experiment with distractor objects. The distractor objects were not in the training data and were placed in physical space during rollout.

C. Method 2: AV-History

In this second method, we use the acoustic video as a reactive policy input during deployment. We feed the temporally accumulated “history” map, the per-pixel maximum over all frames up to the current one (Fig. 3a, History). The blue-colored components in Fig. 3 show the integration of AV-History into the policy architecture.

The 50×90 input history map is resized and average-pooled onto the same 20×35 patch grid as the overhead image, yielding one scalar per patch. The AV history map does not pass through the DINO encoder. Instead, each scalar is embedded by a lightweight per-patch, two-layer MLP. The AV history already shares the same view geometry as the overhead image. To leverage this known spatial correspondence without learning it through attention, we apply element-wise addition of AV history features to the corresponding overhead DINO token, instead of concatenating it as additional tokens. The token count therefore is identical to the Vision-Only model, which adds no additional attention cost when the acoustic-video modality is incorporated.

VI. EXPERIMENTS

To study the efficacy of our proposed perception system, we evaluated imitation learning policies on five vibroacoustic manipulation tasks in real environments, compared to two baseline policies. We ran 20 trials per task, which yielded 400 evaluation trials in total (4 policies $\times$ 5 tasks $\times$ 20 trials).

A. Vibroacoustic Manipulation Tasks

We focus on tasks involving vibroacoustic sound that is naturally induced by friction during manipulation.

Task 1: Cable Threading. The robot routes a USB-A cable through a hollow conduit until the connector exits, then grasps it to pull it through the conduit (Fig. 4 *Cable Threading*). The cable is visually occluded inside the raceway, and spatial audio provides internal contact state. A trial is considered successful if the robot pulls it successfully.

Task 2: Plastic Grinding. The robot picks up a 3D-printed part and brings it to a rotary filing tool (Dremel) to abrade a 1 cm blue mark until the majority of the mark is removed (Fig. 4 *Plastic Grinding*). Acoustic emission from the grinding contact provides an explicit contact cue. A trial is considered successful if the robot abrades the majority (i.e., 50%) of the blue mark.

Task 3: Transparent Film Peeling. The robot peels a transparent protective film from a smartphone screen (Fig. 4 *Transparent Film Peeling*). The policy should detect failure under manipulation of transparent material and re-attempt grasping based on spatial acoustic feedback. A trial is considered successful if the film is fully detached from the screen.

Task 4: Sealed-Box Opening. The robot grasps a utility cutter, slices through a 9 cm tape seal on a box, and opens the top flaps (Fig. 4 *Sealed-Box Opening*). The acoustic signal confirms continuous cutting contact across the seam. A trial is considered successful if the tape is fully cut and the box is unsealed.

Task 5: Surface Cleaning. The robot wipes a 29 cm target mark off an industrial handle (Fig. 4 *Surface Cleaning*). Specular glare on the metallic surface impairs visual tracking. A trial is considered successful if the robot wipes off $>67\%$ of the blue mark.

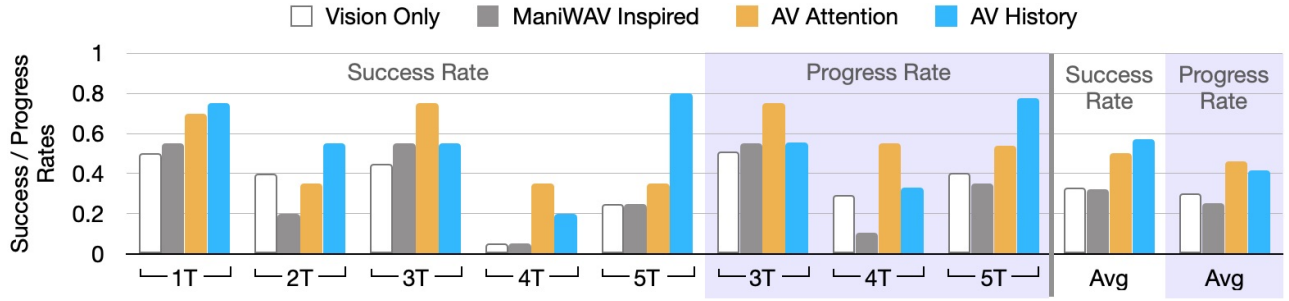


Fig. 5. End-to-end success on real tasks. Each task is evaluated on 20 trials for each policy, which yields 400 evaluation trials. In addition to success rate, we report progress rate for the three tasks in which continuous execution (e.g., cutting the entire seal) is required for success.

B. Baselines

We evaluate 2 baseline approaches: Vision-Only and ManiWAV-Inspired. To ensure a fair comparison, all baseline policies are trained and evaluated using identical datasets, hyperparameter settings, and deployment parameters (1 second action chunk size, 20 Hz inference frequency, and exponential temporal aggregation decay of 0.05) as our proposed methods. The only differences are in acoustic-video observations and the corresponding architecture described in Fig. 3 (blue, orange). We have two baseline policies, and the description of each architecture follows:

1) Vision Only evaluates policy performance using visual observations from three camera streams (overhead and two wrist views) in Fig. 3 (gray).

2) ManiWAV Inspired adapts an approach from ManiWAV [28], where a single-channel audio spectrogram is fed into the policy as an input. We use only channel 1 (center-left) microphone from the array. We follow the subsequent implementation and parameters from ManiWAV [28]. The microphone we use has a native sample rate of 44.1 kHz, therefore we downsample the audio to 16 kHz. Similarly, we extract a 64-bin log-mel spectrogram using a 25 ms Hann window with a 10 ms hop length following [28]. At each time step, a 2.0 s history window of the spectrogram is processed by a learned CNN encoder into 8 audio tokens. These tokens are then concatenated with the proprioception, visual, and action query tokens, resulting in 1039 total tokens for transformer self-attention. Following [28], we apply ESC-50 environmental audio augmentations and an audio spectrogram dropout rate of 0.5 during training. We found both steps crucial to prevent the policy from overfitting to the audio modality. Yet, we acknowledge that this baseline is inspired by ManiWAV’s approach rather than an exact implementation, because we do not use a contact microphone or a diffusion UNet.

We compare these two baselines to our two proposed methods, which yield four policy models for evaluation per task.

C. Policy Training & Deployment

We collected 40 demonstrations for each task using a teleoperation setup. We train policies with an action chunk size of 1.0 s on NVIDIA GeForce RTX 5090 GPU. During

rollout, the policy runs inference at 20 Hz (every 50 ms) and applies temporal aggregation across overlapping action chunks using an exponential decay factor of 0.05. All model inference is executed on an Ubuntu 24.04 LTS desktop with the same GPU machine, processing real-time data streams from three cameras and the 16-channel microphone array.

D. Real Rollout Environments

For real-world evaluation, we execute 20 rollout trials per method for each task. Each trial is terminated either upon successful task completion or when reaching a maximum timeout of 2 minutes, roughly $2\times$ of averaged teleoperated demonstration length.

To understand whether acoustic-video perception helps the policy to learn task-related features better, we added visual distractors (i.e., objects that were not seen in the training dataset) around the task environment. This includes boxes, screwdrivers, cutters, gloves, and a cleaning liquid bottle, etc (zero to three distractors per scene). We also varied the task object position by 5-10 cm displacement. The 20 evaluation environments for each task are visualized in the last row of Fig. 4 *Eval Env*. For a fair comparison between methods, placed both task objects and visual distractors in the same configuration across methods by comparing the first frame of the compared camera views for each rollout condition.

E. Robot Hardware & Data Rate

We demonstrate Acoustic-video perception on a bimanual robot platform built from two I2RT YAM arms [60] with linear grippers, controlled at 50 Hz. Each arm has an Intel RealSense D405 wrist camera streaming 640×480 image at 30 FPS. The acoustic image is computed from 2048-sample blocks of the 44.1 kHz, giving an effective beamforming rate of 21.5 Hz. For training, camera images and AV input are both resampled onto the 50 Hz robot control clock, so every policy step sees a temporally aligned acoustic-video observation.

F. Evaluation Metrics

The binary success criteria for each task are described in Sec. VI-A. Additionally, we report the task progress rate to evaluate partial performance across tasks 3T, 4T, and 5T. Progress measures the fraction of task completion per trial. For instance, in *Surface Cleaning*, task progress is defined as

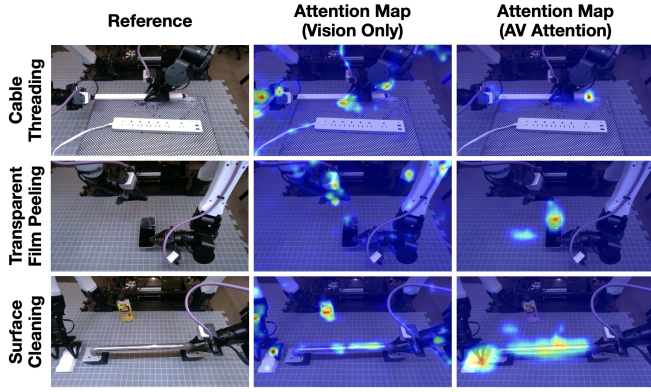


Fig. 6. Action-query-to-image-patch attention map overlaid on the input frame. Middle: Without AV-Attention, the policy spreads attention over background clutter (See *Surface Cleaning*). Right: Models with AV-Attention better attend to task-relevant pixels, e.g., attending to the USB plug tip in *Cable Threading* or even to transparent materials better in *Transparent Film Peeling*.

$S_{\text{progress}} = \frac{B_{\text{grasped}}}{B_{\text{initial}}}$ where B is a blue mark and its initial length is 29 cm. Similarly, task progress for *Transparent Film Peeling* and *Sealed-Box Opening* is defined as the normalized length of detached film and cut tape, respectively, relative to their initial total lengths.

VII. RESULTS

The overview of the experiment results is shown in Fig. 5. Across five real-world tasks, acoustic-video perception increases task performance compared to both Vision-Only and ManiWAV-inspired baselines.

1T: Cable Threading Task. The primary failure mode is exit detection ambiguity (Fig. 7, *Cable Threading*), causing premature grasping or continuous over-pushing due to exit-timing misestimation (success rates Vision-Only: 0.5, ManiWAV-Inspired: 0.55). Both proposed AV methods reduce failures.

The AV-Attention method increases the success rate from 0.5 to 0.7. The policy with AV-Attention better allocates the spatial attention weights to task-relevant visual features, such as the conduit aperture at the critical instant of exit. As shown in Fig. 6 (*Cable Threading*), we found that the Vision-Only baseline diffuses attention across background regions during cable exit, while our model attends to the emerging cable.

The AV-History approach achieves the highest performance in this task, increasing the success rate to 0.75. The explicit auxiliary input about the cable location may help the policy to track progression robustly even while the cable remains visually occluded most of the time.

Other failure types we observed include failing to grasp the cable after accurate exit detection and accidentally grasping the bottom cover together with the cable.

2T: Plastic Grinding Task. AV-History achieved the highest success rate (0.55), outperforming both the Vision-Only (0.40) and ManiWAV-Inspired (0.20) baselines. The ManiWAV-Inspired policy was especially prone to this. Both baselines frequently ground the wrong location on the part, whereas both AV policies ground the correct location in most

trials, suggesting that Acoustic Video aids spatial alignment of the part against the tool.

The dominant failure across all policies was failing to make contact between the part and the grinding bit. We found that the grinding contact is occluded by the tool in the overhead view, so there is little task-relevant visual evidence of contact for AV-Attention to re-weight (success rate 0.35). AV-History, by contrast, uses the contact location as a reactive observation rather than re-weighting existing pixels, which shows better performance than AV-Attention.

Remaining failure modes include the assistive arm failing to grasp the plastic.

3T: Transparent Film Peeling Task. In this task, baseline models struggle during film grasping of the transparent film, including missed grasp attempts, slippage from an unstable grasp, and a lack of retry behavior following an unsuccessful grasp. As a result, the progress rates of Vision-Only and ManiWAV-Inspired were 0.51 and 0.55, respectively.

In contrast, the policy with the AV-Attention method reduces grasp-related errors, increasing the progress rate to 0.75. We also observed in the attention visualization that the policy weights subtle features such as specular film edges better (Fig. 6, *Transparent Film Peeling*). The AV-History method also increases the progress rate to 0.56, leveraging spatial acoustic feedback to track contact state during peeling and recovering when non-film contact is detected. We observed that AV-History sometimes failed at the initial stage of grasping the phone when distractor objects were present. This suggests that AV-Attention, which focuses visual attention on task-relevant regions, may yield more consistent behavior under visual clutter than AV-History.

Other failure modes include tipping the device over, which makes the policy unable to proceed further.

4T: Sealed-Box Opening Task. Baselines achieve success rates of 0.05 for both Vision-Only and ManiWAV-Inspired, respectively. A common failure mode in baselines is the policy prematurely terminating the cutting stage (i.e., putting down the cutter) and attempting to open the flaps before the seal is fully cut. Both AV approaches demonstrated improved detection of box cut status, improving success rate to 0.35 for RGBS-Attention and 0.2 for RGBS-History.

The second failure mode is around the cutting precision. Baseline policies frequently misaim toward the seam of box flaps, resulting in only partial tape cuts (progress rate Vision-Only: 0.29, ManiWAV-Inspired: 0.11). In contrast, the AV-Attention policy often shows an explicit aiming stage and achieves more complete cuts (0.55), while the AV-History policy shows marginal improvement (0.33).

Additional failure modes include applying insufficient blade pressure to cut the tape or excessive pressure that damages the box. We also note several cases classified as failures in which only a sub-millimeter uncut seal prevented the flap from opening (Fig. 7 *Box Decanting*), occurring in 3 trials for AV-Attention and 2 for AV-History.

5T: Surface Cleaning Task. Severe specular glare obscures the target blue mark in the *Surface Cleaning* task. As a result, the dominant failure mode is premature task

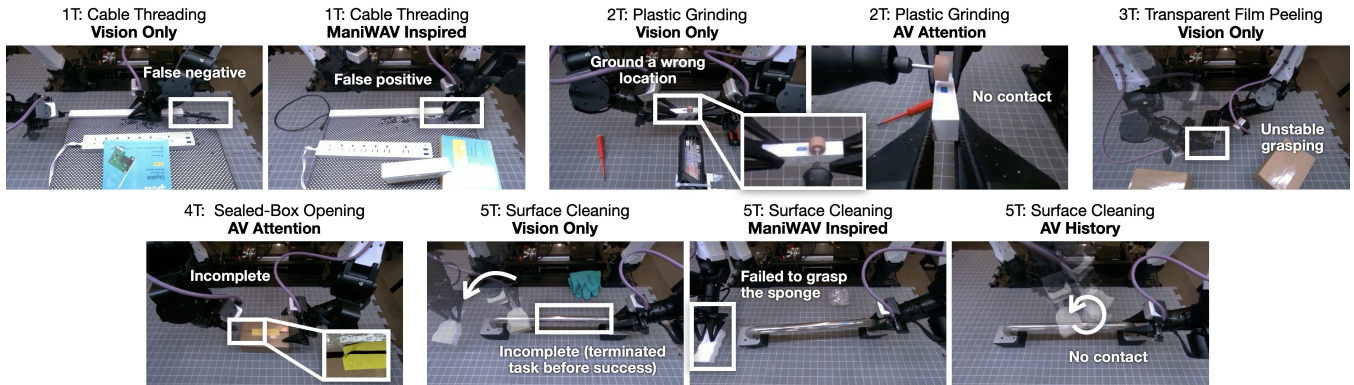


Fig. 7. Failure modes observed during evaluation.

termination following incomplete surface wiping (Fig. 7, *Surface Cleaning*) across Vision-Only baseline (success rate: 0.25), ManiWAV-Inspired baseline (0.25), and AV-Attention model (0.35).

Our AV-History method increases the success rate to 0.80 using acoustic video as an explicit time-varying input. The policy tracks the wiped coverage, substantially boosting task completion.

The guided attention method also shows partial wiping failures, but it achieves higher overall coverage (progress rate: 0.53) than the baselines (0.38). Attention maps in Fig. 6 (*Surface Cleaning*) demonstrate that acoustic regularization directs visual focus toward the wiped surface rather than visual distractors, improving the blue mark detection despite high surface reflectivity.

In addition, baseline models struggle to pick up the sponge under visual distractors or extreme initial poses, whereas neither AV method exhibits this failure. Unlike many grasps, which are acoustically silent, compressing the porous sponge produces a distinctive sound; this gives the AV models a reliable contact cue that vision alone does not provide.

Other failure modes are that the policy is stuck in a wiping motion without making contact, and the robot drops the target object, which makes the policy unable to proceed with the task further.

VIII. LIMITATION & FUTURE WORK

Although we applied acoustic filters to remove ambient roomtone noise (e.g., typical mechanical vibration, AC-driven lighting, etc., whose fundamental frequencies span mostly < 100 Hz), our pipeline does not consider other types of sound distractors with harmonic frequencies (e.g., music, human speech, etc). Because our target contact sounds are at 10 kHz (Sec. IV-B), such distractors are partially attenuated; however, their harmonics can still produce spurious activations in the acoustic video. Addressing this robustly is a promising direction for future work, for instance, through spatial masking of fixed distractor locations or using a source classifier.

We currently supervise attention over the entire remaining future horizon. While we found this to be more effective

than supervising with only the current single frame of acoustic-video observation, the entire future horizon may be unnecessary. Using supervision to a fixed future time window or contextual-level window needs further exploration.

As we defined our problem setup in Sec. III, our current implementation around the AV-History approach performs no dynamic tracking. Extending it to handle moving objects remains future work.

Some commercial robots already ship with built-in microphone arrays (e.g., Unitree G1 [41], R1 [42]). However, such arrays have been used for sound-source separation in service of human-robot voice interaction (i.e., isolating a human speaker in a noisy environment [61], [62]), and none have been applied for visually grounded acoustic sensing and manipulation. Extending our approach to these platforms is a promising direction, as the sensing hardware is already in place.

IX. CONCLUSION

We introduced acoustic-video as a new perceptual modality for robot manipulation. Using microphone array beamforming, our system localizes contact sound during manipulation and obtains a spatially grounded acoustic map. We presented two methods for incorporating it into imitation learning policies: AV-Attention, a training-time regularizer that steers visual attention toward contact regions, and AV-History, an accumulated acoustic map as a reactive policy observation. Across five vibroacoustic tasks in real environments, acoustic-video perception improved task performance compared to two baselines. We hope this work encourages further use of spatially grounded acoustic sensing in robot learning.

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